Business Problem

In recent years, Hotels often face booking cancellations, which leads to revenue loss and inefficient room management. This project analyzes hotel booking data to identify key factors that influence cancellations and provide insights to help hotels reduce cancellation rates.

Assumptions

1. The data is relevant and useful for analysing hotel business decisions.
2. There are no negative effects if hotels apply the suggested strategies.
3. The hotels are not currently using the recommended solutions.
4. Booking cancellations are the main factor affecting hotel revenue.
5. When bookings are cancelled, rooms remain empty during that period.
6. Customers usually book and cancel reservations within the same year.

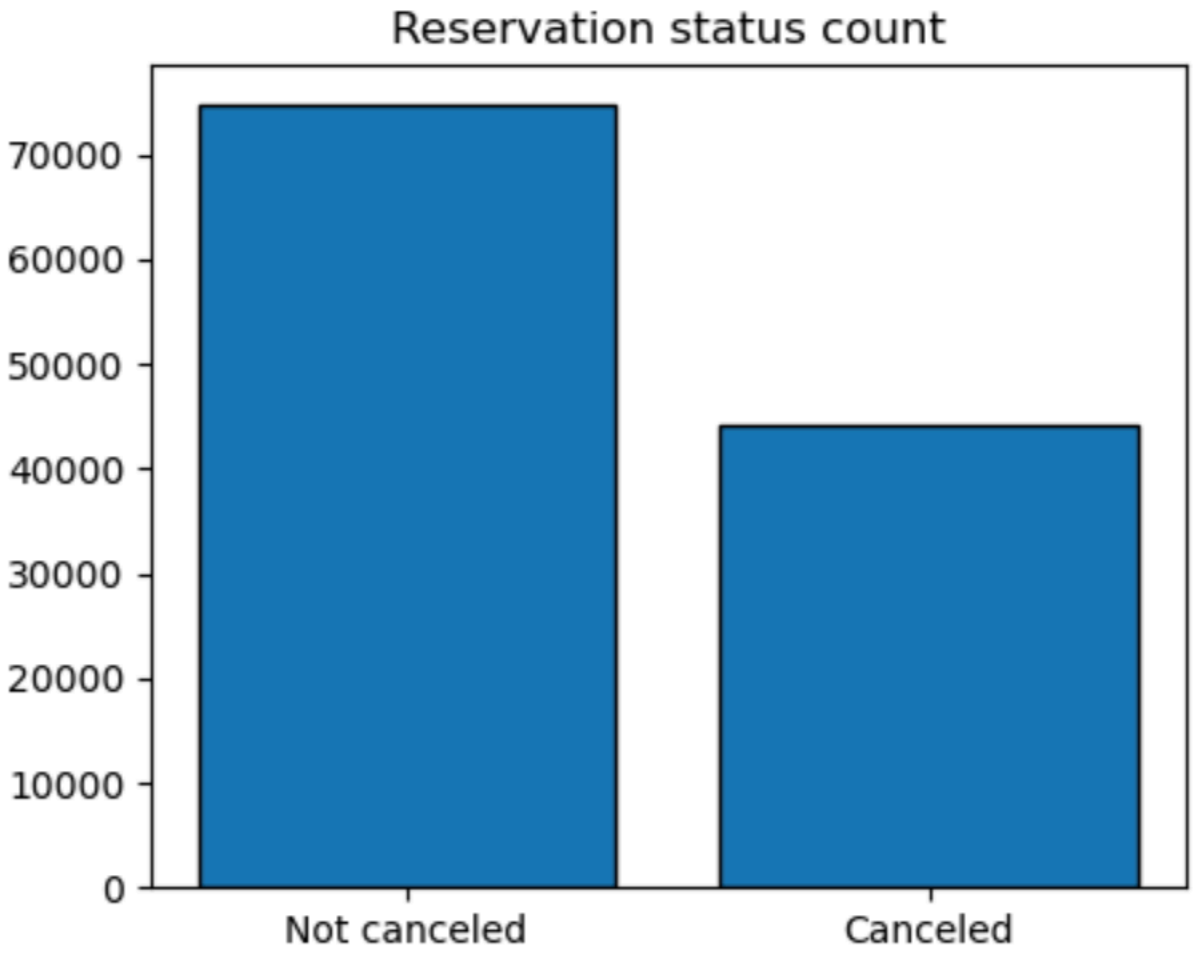
Research Question

1. What are the variables that affect hotel reservation cancellations?
2. How can we make hotel reservations cancellations better?
3. How will hotels be assisted in making pricing and promotional decisions?

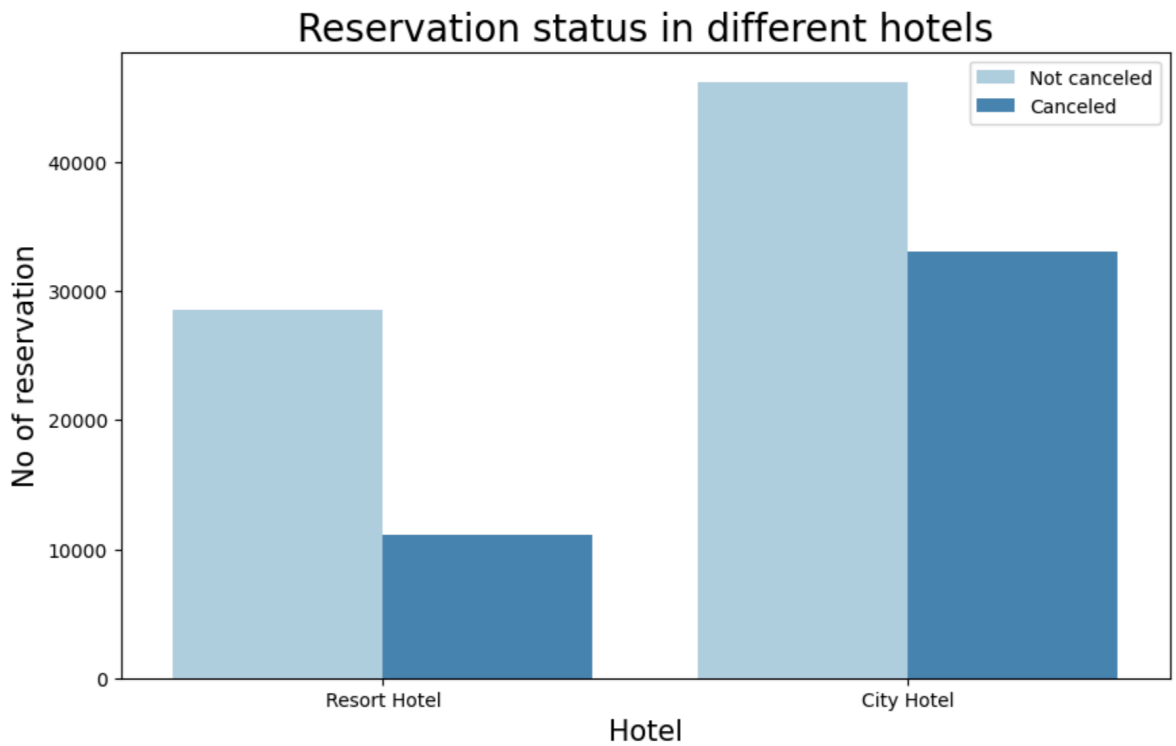
Hypothesis

1. More cancellations occur when prices are higher.
2. When there is a longer waiting list, customers tend to cancel more frequently.
3. The majority of clients are coming from online travel agents to make their reservations.

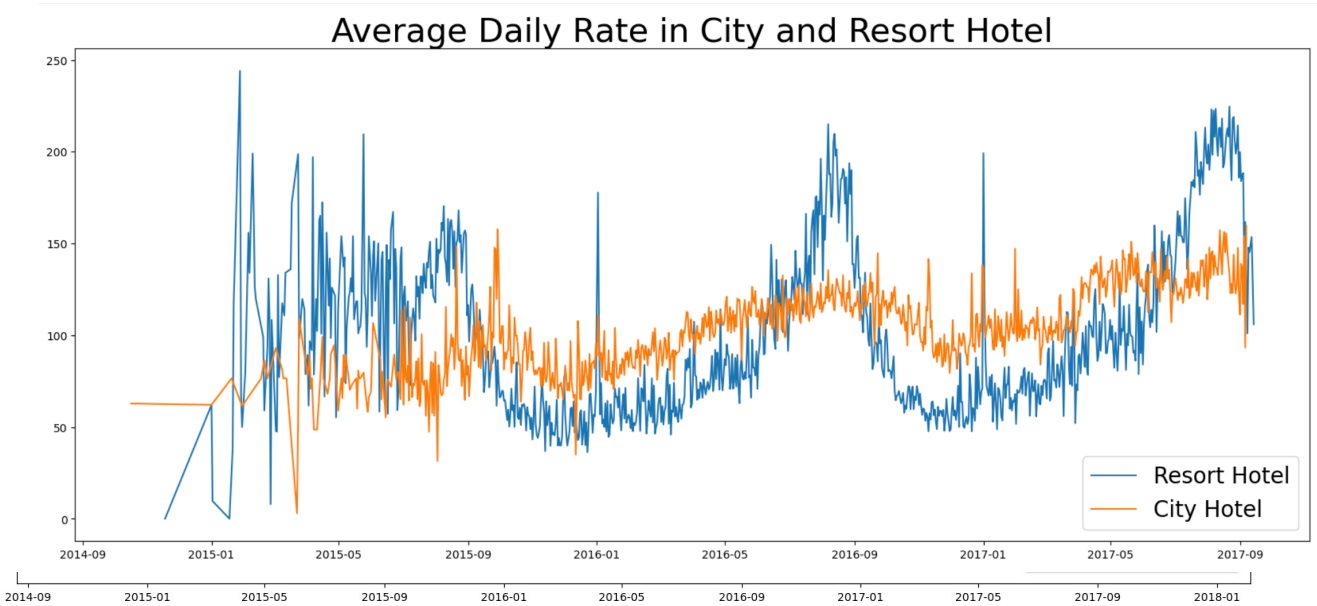
Analysis and Findings



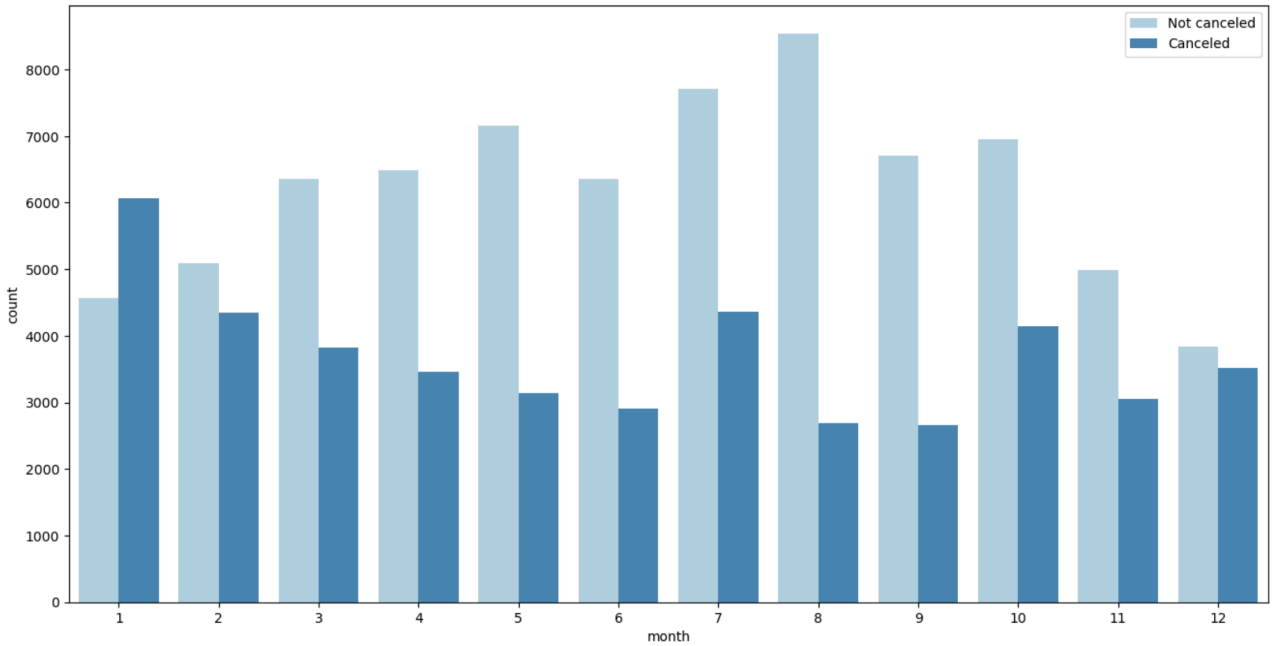
The bar graph shows the percentage of canceled and non-canceled reservations. Most reservations were not canceled, but 37% of customers canceled their bookings. This high cancellation rate negatively affects the hotel's revenue.



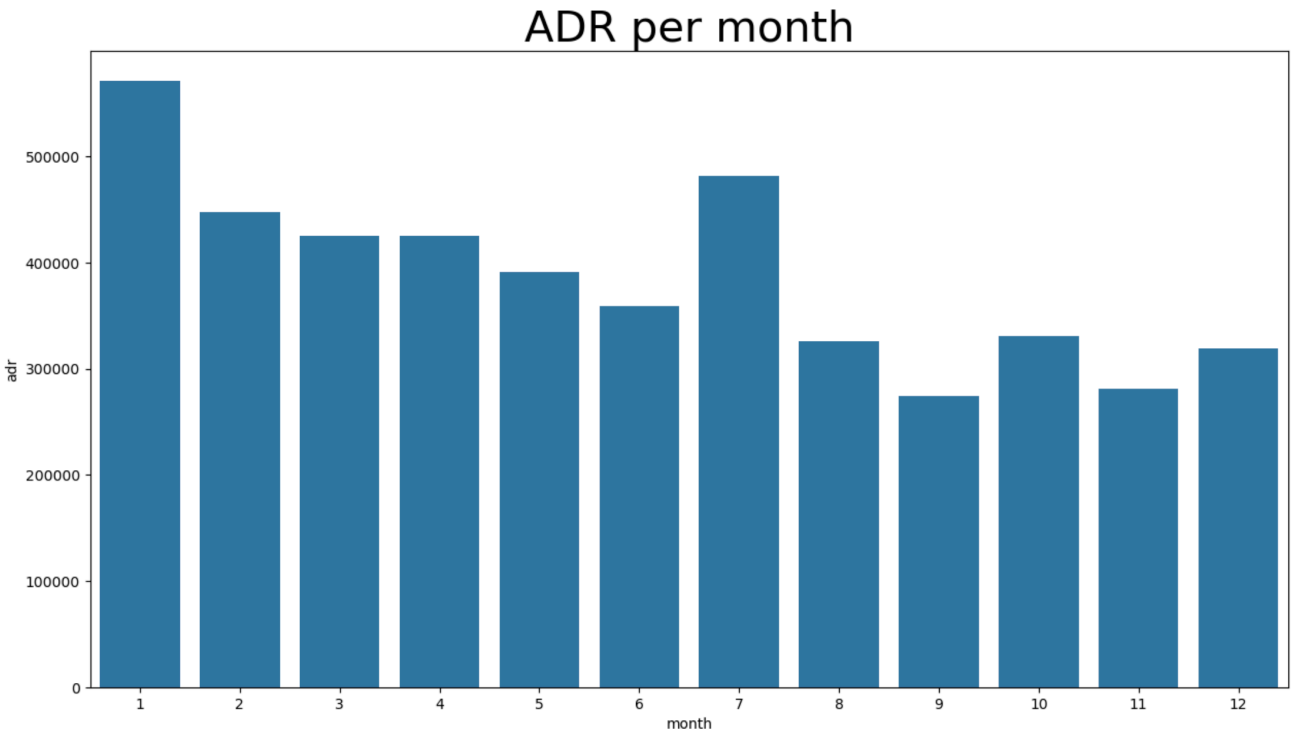
In comparison to resort hotels, city hotels have more bookings. It's possible that resort hotels are more expensive than those in cities.



The line graph above shows that, on certain days, the average daily rate for a city hotel is less than that of a resort hotel, and on other days, it is even less. It goes without saying that weekends and holidays may see a rise in resort hotel rates.

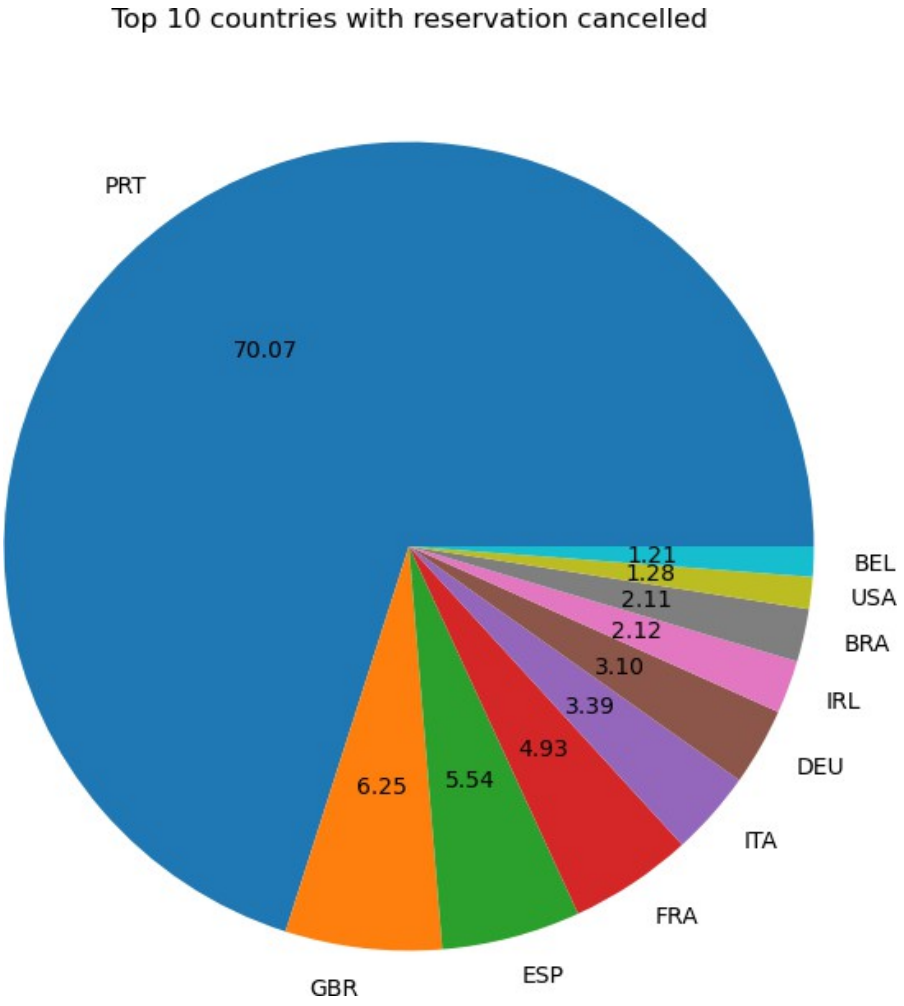


The grouped bar graph shows the number of confirmed and canceled reservations by month. Both confirmed and canceled bookings are highest in August. However, January has the highest number of cancellations.

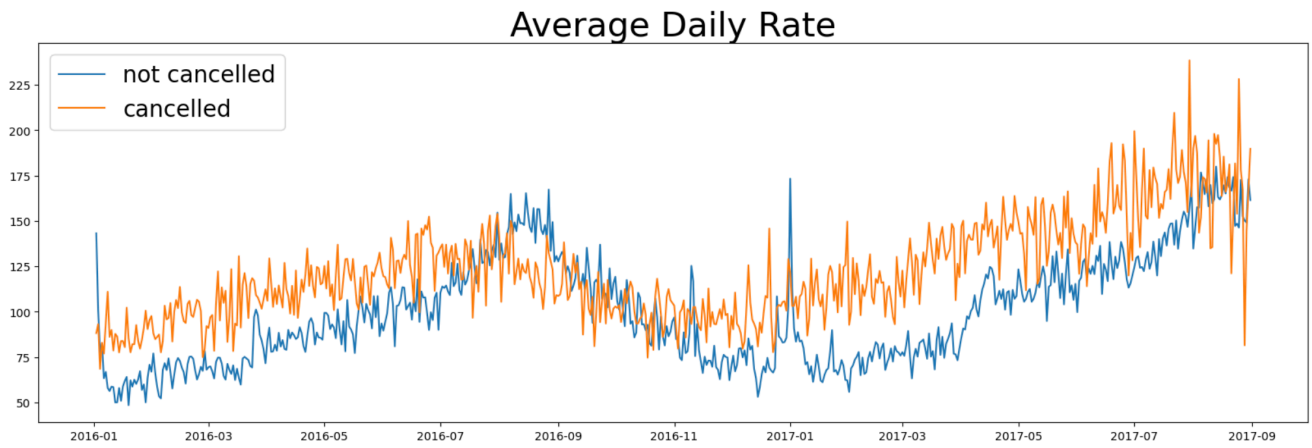


This bar graph demonstrates that cancellations are most common when prices are greatest and are least common when they are lowest. Therefore, the cost of the accommodation is solely responsible for the cancellation.

Now, let's see which country has the highest reservation canceled. The top country is Portugal with the highest number of cancellations.

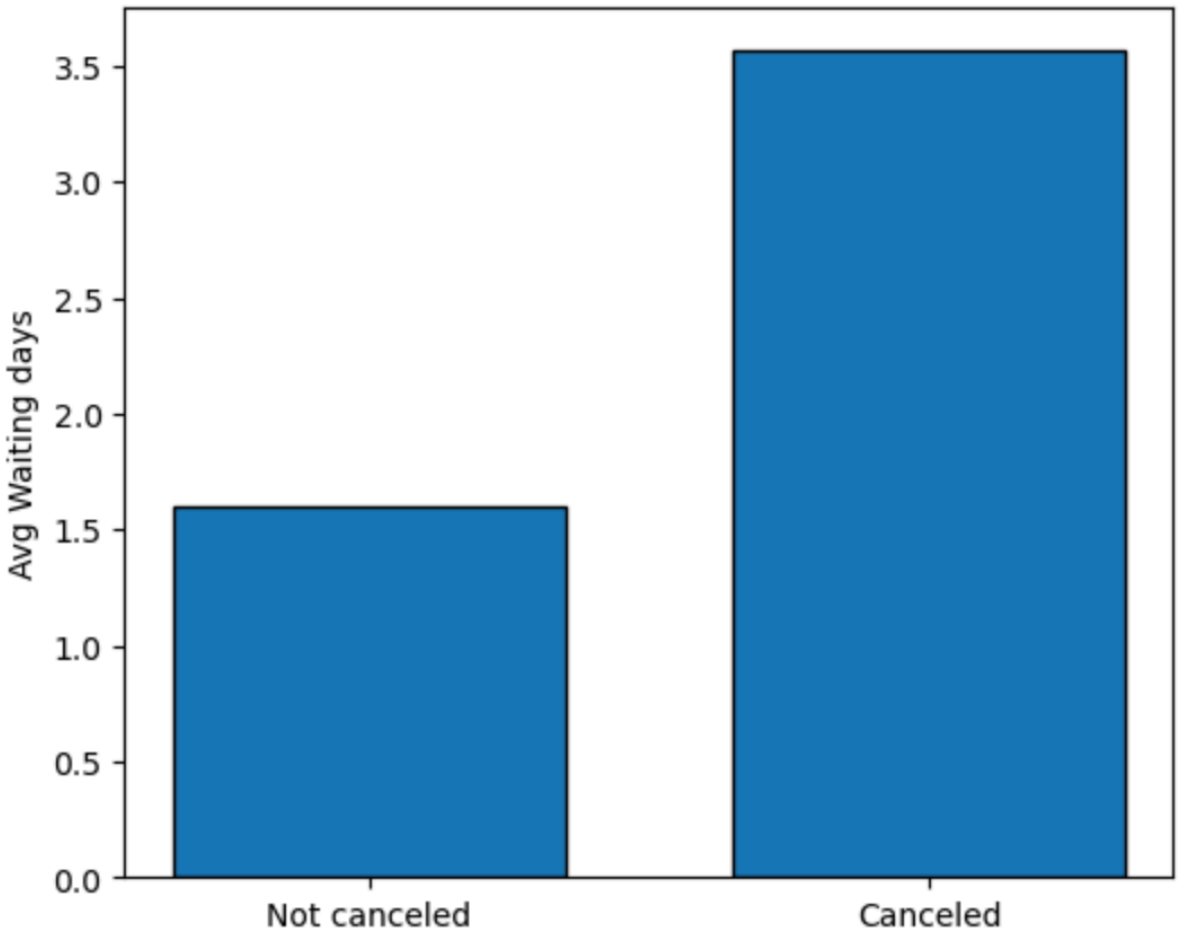


This analysis shows where hotel bookings come from. About 46% of customers book through online travel agencies, and 27% come from group bookings. Only 4% of customers book hotels directly by visiting the hotel.



As seen in the graph, reservations are canceled when the average daily rate is higher than when it is not canceled. It clearly proves all the above analysis, that the higher price leads to higher cancellation.

The analysis shows that bookings with longer waiting days have a higher chance of cancellation. The average waiting days for canceled bookings is 3.57, while for non-canceled bookings it is only 1.60. This indicates that customers who wait longer are more likely to cancel their reservations.



Suggestions

1. Cancellation increases when room prices are high. Hotels should improve their pricing strategy and offer discounts to reduce cancellations.
2. Resort hotels have more cancellations than city hotels. They can offer discounts on weekends or holidays to attract more customers.
3. January has the highest cancellations. Hotels should run marketing campaigns or promotions during this month to increase bookings.
4. Hotels in Portugal should improve their service quality to reduce cancellations.
5. Hotels should reduce waiting days for booking confirmation to improve customer satisfaction and reduce cancellations.